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NATURAL LANGUAGE PROCESSING (CO3085)

Assignment

"Duplicate question detection"

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2.1 Problem description

In many online question forum, such as Quora, StackOverflow, or Reddit, where people post a question and other people post their answer, it's common that it's a question that has been asked before with different wordings. For example, 'What is Segmentation fault?' is the same as 'Can somebody please explain to me what segfault is?? I keep getting this error.' It would be helpful to show the asker the answers in the similar post, to not waste time trying to answer the same question over and over again.

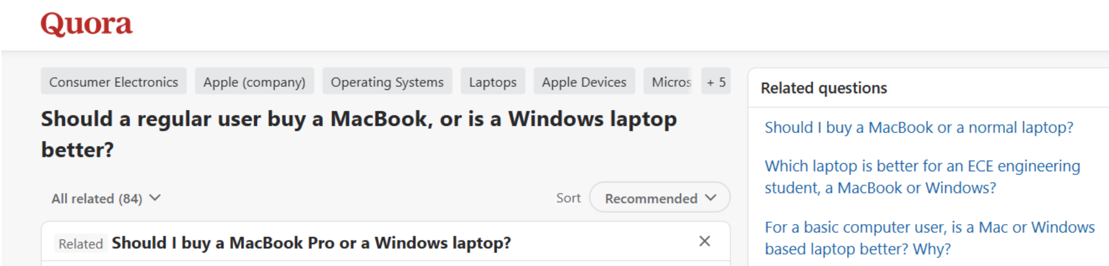

Figure 1: Quora related questions

This is a problem of 'Semantic similarity', in this case it's specifically 'duplicate question detection'. We are trying to determine whether 2 question has the same, or close enough, sematic meaning. In the scope of this project, we will work with the dataset 'Quora question pairs', with each sample containing 2 questions and a label telling whether those 2 are duplicated or not.

Historically, processing natural language has always been a complex and tedious task with countless rules and contraints, as languages are always evolving and new words, phrases, meaning keep changing over time, so much that ordinary grammar rules struggle to keep up with. In recent years, thanks to the development of computing and modern deep learning architecture, especially Transformer, the approach to these problem has been more friendly, as we move into extracting embedding vector to represent the meaning of natural language. In this project, we will be looking at the 2 commonly used model for this task, which is BERT and SBERT.

2.2 Dataset EDA

First, we need to understand the dataset we are about to work with.

We download the dataset from Kaggle, then use Python library `pandas` to load the csv file to memory.

We try printing the first 10 rows from the dataset, shown in table 1. The important bits are the text of question 1, question 2, and the label indicating whether they are duplicated. Two sentence may look similar but can mean entirely different things.

After that, we clean the null values from the table, and display some information about the table. We see the dataset has 404348 samples, with 149306 positive and 255042 negative samples. This mean the dataset is quite imbalance, hence during evaluation we will use metrics like precision, recall, f1 score, together with accuracy.

We also do some basic plotting, with the histogram of character counts and words count, plus the word cloud showing the most appearing words in our dataset.

ID	QID1	QID2	Question 1	Question 2	Duplicate
0	1	2	What is the step by step guide to invest in share market in India?	What is the step by step guide to invest in share market?	0
1	3	4	What is the story of Kohinoor (Koh-i-Noor) Diamond?	What would happen if the Indian government stole the Kohinoor (Koh-i-Noor) diamond back?	0
2	5	6	How can I increase the speed of my internet connection while using a VPN?	How can Internet speed be increased by hacking through DNS?	0
3	7	8	Why am I mentally very lonely? How can I solve it?	Find the remainder when 23^{24} is divided by 24,23?	0
4	9	10	Which one dissolve in water quickly: sugar, salt, methane and carbon dioxide?	Which fish would survive in salt water?	0
5	11	12	Astrology: I am a Capricorn Sun, Cap moon and Cap rising... what does that say about me?	I'm a triple Capricorn (Sun, Moon and ascendant in Capricorn). What does this say about me?	1
6	13	14	Should I buy Tiago?	What keeps children active and far from phone and video games?	0
7	15	16	How can I be a good geologist?	What should I do to be a great geologist?	1
8	17	18	When do you use I instead of l?	When do you use "&" instead of "and"?	0
9	19	20	Motorola (company): Can I hack my Charter Motorola DCX3400?	How do I hack Motorola DCX3400 for free internet?	0

Table 1: Sample records from the Quora Question Pairs dataset.

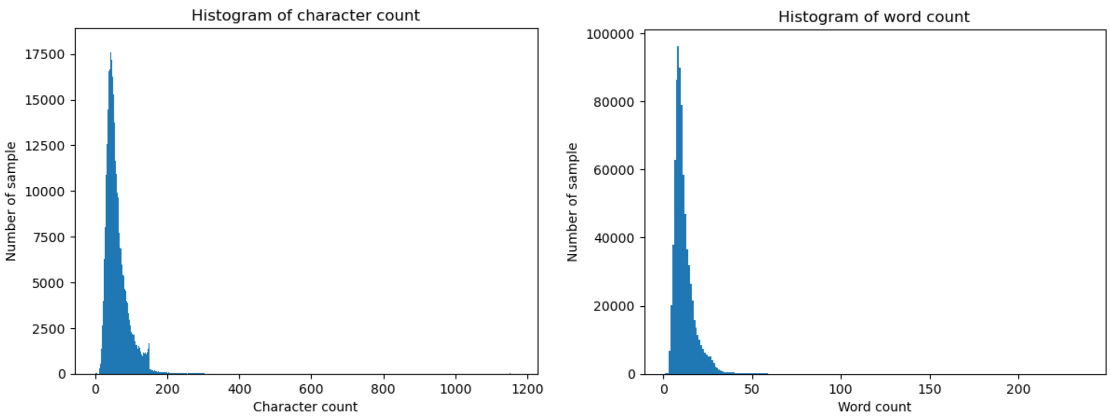

Figure 2: Histograms

Figure 3: Word cloud

3 Literature review

Before we dive into the problem, let's review the 2 important paper that explain how the 2 models we will be using work: BERT and SBERT.

3.1 BERT

BERT (Bidirectional Encoder Representations from Transformers) was introduced in the paper "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" by Devlin et al at Google AI Language in 2019. Unlike other language models, like the original "Attention is all you need" paper (encoder-decoder) or GPT (decoder only), BERT has encoder only architecture, and it was pretrained without future masking, allowing information to flow in 2 direction, left to right and right to left, making the model able to represent better the whole sequence. The pretrained model is then finetuned for various downstream tasks.

There are 2 step in BERT framework: pre-training and fine-tuning. The model is first pre-trained on unlabeled data. The fine-tuning step take the pre-trained BERT model and train the parameters with their respective labeled data.

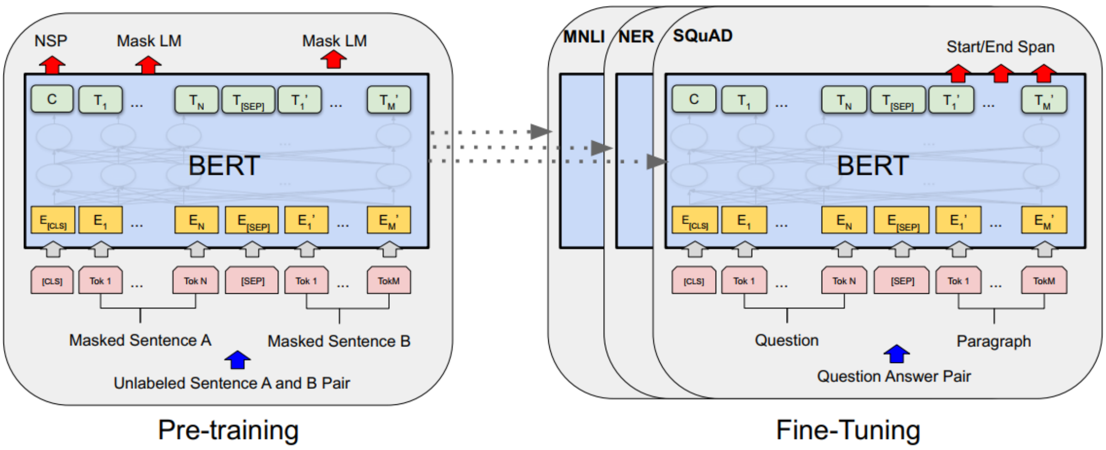

Figure 4: BERT training

BERT's architecture is a multiple transformer encoder layer, with 2 model sizes, BASE (768d hidden size, 110M params) and LARGE (1024d hidden size, 340M params). BERT allow the input to be either a single sentence or a pair of sentence (question-answer, next sentence, etc.). It use WordPiece embedding with vocab size of 30000. The first token is always `CLS` for classification tasks. If the input is a pair of sentences, there will be a `SEP` token between them.

BERT was pre-trained on BooksCorpus (800M words) and English Wikipedia (2500M words), with 2 tasks: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP).

- In MLM, random token are masked and the model has to predict them. Because we are predicting the `MASK` token instead of the next token (like GPT), we can allow attention to flow to both past and future token, allowing bidirectional information
- In NSP, we choose a pair of sentences and have the model predict whether sentence B is the next sentence of sentence A in the corpus. This task help the model understand relationship between sentences and improve downstream task, like Question Answering (QA) and Natural Language Inference (NLI)

For fine-tuning, BERT was trained end-to-end from the pre-trained parameters with the tasks' input and output. At the time it achieved SOTA on 11 tasks:

- General Language Understanding Evaluation (GLUE):
  - MultiNLI Matched/Mismatched (MNLI-(m/mm))
  - **Quora Question Pairs (QQP)**
  - Question NLI (QNLI)
  - The Stanford Sentiment Treebank (SST-2)
  - The Corpus of Linguistic Acceptability (CoLA)
  - Semantic Textual Similarity Benchmark (STS-B)
  - Microsoft Research Paraphrase Corpus (MRPC)
  - Recognizing Textual Entailment (RTE)
- The Stanford Question Answering Dataset (SQuAD)
  - SQuAD v1.1
  - SQuAD v2.0
- The Situations With Adversarial Generations (SWAG)

System	MNLI-(m/mm) 392k	**QQP** 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
$\text{BERT}_{\text{BASE}}$	84.6/83.4	**71.2**	90.5	93.5	52.1	85.8	88.9	66.4	79.6
$\text{BERT}_{\text{LARGE}}$	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

Table 2: GLUE results

Here we focus on QQR, as this will be the dataset we are working with in this project.

3.2 SBERT

4 Training

5 Result and insight

6 Conclusion